MIL-P-47276A(MI)
23 May 1988
Superseding
MIL-P-47276(MI)
17 August 1976

MILITARY SPECIFICATION

PRIMER, BONDING

This specification is approved for use within the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid organic nitrilephenolic resin primer, referred to herein as the primer.

1.2 Part number. The part number shall consist of the letter "B", the basic number of this specification, and a dash number denoting container size.

Example

B47276 - 1*

- 1* — container size
- B47276 — basic specification number

* 1 - pint
2 - quart
3 - gallon

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8040

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-P-1892	Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of

MILITARY

MMM-A-132	Adhesives, Heat Resistant, Airframe Structural, Metal to Metal

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 562	Standard Test Method for Consistency of Paints Using the Stormer Viscometer
ASTM D 1200	Standard Test Method for Viscosity of Paints, Varnishes and Lacquers by Ford Viscosity Cup

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ASTM D 1259 Standard Test Method for Nonvolatile Content of Resin Solutions

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules, and Regulations

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, (see 6.2) a first article sample shall comply with all requirements of this specification.

3.2 Materials. The primer shall consist of a liquid organic nitrile-phenolic resin of such composition that will yield a product conforming to all requirements of this specification.

3.3 Toxicity - flammability. Identify all known toxic and flammable materials in the primer on the unit and shipping containers (see 5.2). Precautions to be observed in the use of this primer shall also be marked on the containers.

3.4 Physical properties.

3.4.1 Solids content. The solids content of the primer shall be 15 ± 5% by weight (see 4.7.2).

3.4.2 Viscosity. The viscosity of the primer shall yield a value of not more than 36 seconds at 27 degrees celsius (C) (80 degrees Fahrenheit (F) (see 4.7.3).

3.4.3 Density. The density of the primer shall be 7.2 ± 0.2 pounds per gallon (see 4.7.4).

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3.4.4 Dilution stability. A mixture of the primer with methyl isobutyl ketone shall ex ibit no curdling, precipitation or separation (see 4.7.5).

3.4.5 Surface finish. The coated panel surface shall have a smooth, fine-grained finish, free from streaks, blisters, built-up edges, and other surface irregularities (see 4.7.6.2).

3.4.6 Film thickness. A single spraycoat of the primer shall dry to a thickness of 0.0001 to 0.0003 inch (see 4.7.7).

3.4.7 Shear strength. Joints which are primed with primer conforming to this specification and bonded with film adhesive conforming to MMM-A-132, type I, class 2 form F, group 3 (0.030 psf) shall show shear strengths as specified in table I (see 4.7.8).

TABLE I. Shear strength requirements

Degree of Temperature	Number of panels tested	Shear Strength, Average, Minimum	Shear, Strength, Individual, Minimum
+24 C(+75F)	4	2500 psi	2200 psi
+82 C(+180F)	4	1250 psi	---
-55 C(-67C)	4	2500 psi	2200 psi

3.5 Workmanship. The components shall be mixed and processed in such a way as to ensure compliance with the requirements of this specification. Freshly-opened containers of the primer shall be free from grit, sediment, and suspended matter (see 4.7.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of

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known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. The test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of required examinations and tests.

4.2 Classification of inspections. The inspections of the MK III electrical test set shall be classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under conditions of room ambient pressure, temperature, and humidity.

4.4 First article inspection. Unless otherwise specified, three containers of primer shall be randomly selected from the first lot manufactured under each contract or purchase order. A sample from each container shall be subjected to each of the quality conformance inspections specified in table II in the order listed. The first article sample shall be taken from a batch of primer which has been produced by the same processes as proposed for all of the primer furnished under the same contract or purchase order.

4.5 Quality conformance inspection. All production lots of the primer offered for acceptance hereunder shall be subjected to the quality conformance inspection. Two containers of primer shall be selected at random from each lot offered for acceptance, and samples therefrom shall be subjected to the tests in table II in the order listed. If a failure occurs, the lot shall be rejected.

TABLE II. Quality conformance inspection

Examination or test	Requirement Paragraph	Method Paragraph
Workmanship	3.5	4.7.1
Solids content	3.4.1	4.7.2
Viscosity	3.4.2	4.7.3
Density	3.4.3	4.7.4
Dilution stability	3.4.4	4.7.4
Surface finish	3.4.5	4.7.6.2
Film thickness	3.4.6	4.7.7
Shear strength	3.4.7	4.7.8

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4.6 Inspection lot. An inspection lot shall be as defined in MIL-STD-105 and shall consist of a single batch of primer offered for inspection at one time. Each batch shall have been produced in a single vat in no more than a 24-hour continuous production period with no change in process or raw materials.

4.7 Test methods.

4.7.1 Visual examination for workmanship. The primer in freshly opened containers shall be visually examined to determine compliance with the requirements specified in 3.5. If necessary, small quantities of the primer shall be poured onto clean glass surfaces for examination. Once opened for inspection, containers shall be used within an 8-hour day or shall be discarded.

4.7.2 Solids content test. The solids content of the primer shall be measured in accordance with ASTM D 1259 to verify compliance with the requirements specified in 3.4.1.

4.7.3 Viscosity test. The viscosity of the primer shall be measured with a Ford Cup Viscosimeter with .134 inch orifice, in accordance with ASTM D 1200, to verify compliance with the requirements specified in 3.4.2.

4.7.4 Density test. The density of the primer shall be measured in accordance with ASTM D 562, to verify compliance with the requirements specified in 3.4.3.

4.7.5 Dilution stability. The dilution stability of the primer shall be determined as follows:

a. Mix 50 ± 1 milliliters of the primer with 50 ± 1 milliliters of methyl isobutyl ketone.

b. Place the mixture in a well-stoppered glass container, and store it for 4 to 4 1/2 hours at 23° ± 2°C (73 ± 4°F).

c. At the end of this period, examine the material for compliance with the requirements specified in 3.4.4.

4.7.6 Surface finish tests.

4.7.6.1 Preparation of test panels. Twelve aluminum test panels shall be prepared as specified in MMM-A-132 with the following exceptions:

a. The aluminum shall be wiped clean, de-greased, cleaned for 10 minutes in a solution containing 39 parts water by weight, 10 parts Baume sulphuric acid and one part sodium dichromate, at a temperature of 60°C to 71°C (140°F to 160°F), rinsed with tap water and then distilled water, and dried at 51°C to 66°C (125°F to 150°F) for 15 to 30 minutes in a forced-air circulation oven.

b. The primer shall be applied by spray gun to the area to be bonded, and after air-drying for 10 to 15 minutes, the primed parts shall be placed in a circulating air type oven at 79 ± 15°C (175 ± 27°F) for 30 minutes.

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c. After the primer has dried, pairs of panels shall be bonded together by applying film adhesive conforming to MMM-A-132 type I, class 2 form F, group 3 (0.030 psf), between the primed surfaces and curing the bonds under a pressure of 150 + 5 pounds per square inch (psi) and a temperature of 160 + 3 - 5°C (320 + 5 - 10°F) for 180 ± 5 minutes.

4.7.6.2 Visual examination for surface finish. After the panels have been cured, they shall be visually inspected for compliance with the surface finish requirements specified in 3.4.5.

4.7.7 Film thickness test. The thickness of the film on each test panel shall be measured to verify compliance with the requirements specified in 3.4.6.

4.7.8 Shear strength test. The test panels used for visual inspection shall be subjected to the shear strength tests specified in MMM-A-132, ten minutes after reaching equilibrium at the required temperature, to verify compliance with the requirements specified in table I.

4.7.9 Toxicity certification. The supplier shall certify that the primer complies with the requirements specified in 3.3. At the option of the procuring activity, the primer may be tested to determine whether non-permitted chemicals are included in the material.

4.8 Packaging inspection. Each lot offered for acceptance shall be subjected to packaging inspection to determine compliance with paragraphs 5.1 and 5.2.

4.8.1 Sampling and testing. Samples shall be taken in accordance with the applicable specifications and standards referenced in 5.1 and 5.2. When no sampling plan is specified in those specifications and standards, samples shall be taken in accordance with MIL-STD-105, AQL2.5.

4.8.2 Action in case of failure. Unless other criteria are defined in the applicable documents referenced in 5.1 and 5.2 inclusive, failure of one specimen shall be charged as a single defective. If the quantity of defectives exceeds that permitted, the lot represented by the sample shall be rejected. Defective units shall not be delivered on contract or purchase order until all deficiencies have been corrected.

5. PACKAGING

5.1 Preservation, packing, unitization, and marking. Preservation, packing, unitization, and marking shall be Level A or B (see 6.2).

5.1.1 Level A or B. Preservation, packing, and unitization shall be in accordance with PPP-P-1892.

5.2 Marking. Marking shall be in accordance with MIL-STD-129 and PPP-P-1892. Additional markings shall include the following:

a. Name of the contents: Bonding Primer

b. Title, number, and date of this specification

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c. Manufacturer's name or trademark

d. Manufacturer's CAGE number

e. Batch number

f. Date of manufacturer

g. Quantity

KEEP FROM FREEZING
STORE AT 32 DEGREES F TO 50 DEGREES F

6. NOTES

6.1 Intended use. The primer covered by this specification is intended for use with an adhesive tape which will be pressed between two metal surfaces primed as specified herein.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether first article inspection is required (see 3.1).

c. Level of packaging and packing required (see 5.1).

d. Required warranty of shelf life, if any. Terms such as the following are suggested:

"The supplier shall warrant that the primer will meet all requirements hereof for six months of storage in the unopened container at temperatures from 0° to 10°C (32° to 50°F), and that he will replace any container of primer which fails to meet all requirements hereof during such term of storage.

6.3 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable.

6.4 Subject term (key word) listing.

Primer
Primer, Bonding

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-47276A(MI)	Primer, Bonding

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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